

Mathematical Foundations of Machine Learning

Assignment 9 Solutions

Notation. For an event A in the sample space Ω , we write $\bar{A} = \Omega \setminus A$ for the complement of A .

Question 1 (15 points)

Let X be a continuous random variable with probability density function

$$f_X(x) = \begin{cases} cx(2-x), & 0 < x < 2, \\ 0, & \text{otherwise,} \end{cases}$$

where $c > 0$ is a normalizing constant.

- (a) (4 points) Determine the value of c so that f_X is a valid pdf.
- (b) (6 points) Compute the cumulative distribution function $F_X(x)$ for all $x \in \mathbb{R}$.
- (c) (5 points) Compute $P(X > 1)$ and $P(X > 1 \mid X > \frac{1}{2})$.

Solution.

- (a) For f_X to be a valid pdf, we need

$$\int_{-\infty}^{\infty} f_X(x) dx = 1.$$

Since the density is nonzero only on $(0, 2)$,

$$\begin{aligned} 1 &= \int_0^2 cx(2-x) dx \\ &= c \int_0^2 (2x - x^2) dx \\ &= c \left[x^2 - \frac{x^3}{3} \right]_0^2 \\ &= c \left(4 - \frac{8}{3} \right) \\ &= c \cdot \frac{4}{3}. \end{aligned}$$

Hence

$$\boxed{c = \frac{3}{4}}.$$

- (b) The cumulative distribution function is

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt.$$

Therefore,

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ \int_0^x \frac{3}{4}t(2-t) dt, & 0 < x < 2, \\ 1, & x \geq 2. \end{cases}$$

For $0 < x < 2$,

$$\begin{aligned} F_X(x) &= \frac{3}{4} \int_0^x (2t - t^2) dt \\ &= \frac{3}{4} \left[t^2 - \frac{t^3}{3} \right]_0^x \\ &= \frac{3x^2}{4} - \frac{x^3}{4}. \end{aligned}$$

Thus

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ \frac{3x^2}{4} - \frac{x^3}{4}, & 0 < x < 2, \\ 1, & x \geq 2. \end{cases}$$

(c) First,

$$P(X > 1) = 1 - F_X(1) = 1 - \left(\frac{3}{4} - \frac{1}{4} \right) = \frac{1}{2}.$$

Next,

$$P\left(X > 1 \mid X > \frac{1}{2}\right) = \frac{P(X > 1)}{P(X > \frac{1}{2})},$$

since the event $\{X > 1\}$ is contained in $\{X > \frac{1}{2}\}$. Compute

$$F_X\left(\frac{1}{2}\right) = \frac{3}{4} \left(\frac{1}{2}\right)^2 - \frac{1}{4} \left(\frac{1}{2}\right)^3 = \frac{3}{16} - \frac{1}{32} = \frac{5}{32},$$

so

$$P\left(X > \frac{1}{2}\right) = 1 - \frac{5}{32} = \frac{27}{32}.$$

Therefore,

$$P\left(X > 1 \mid X > \frac{1}{2}\right) = \frac{\frac{1}{2}}{\frac{27}{32}} = \frac{16}{27}.$$

Hence

$$\boxed{P(X > 1) = \frac{1}{2}}, \quad \boxed{P\left(X > 1 \mid X > \frac{1}{2}\right) = \frac{16}{27}}.$$

Question 2 (20 points)

Let X be a continuous random variable with pdf

$$f_X(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Define the events

$$A_1 = \left\{0 < X \leq \frac{1}{2}\right\}, \quad A_2 = \left\{\frac{1}{2} < X < 1\right\}.$$

Suppose another event B satisfies

$$P(B | A_1) = \frac{1}{4}, \quad P(B | A_2) = \frac{3}{4}.$$

- (a) (5 points) Compute $P(A_1)$ and $P(A_2)$.
- (b) (5 points) Use the total probability theorem to compute $P(B)$.
- (c) (5 points) Use Bayes' theorem to compute $P(A_2 | B)$.
- (d) (5 points) Compute $P(A_2 | \bar{B})$.

Solution.

- (a) First compute the CDF of X on $0 < x < 1$:

$$F_X(x) = \int_0^x 2t \, dt = x^2.$$

Therefore,

$$P(A_1) = P\left(0 < X \leq \frac{1}{2}\right) = F_X\left(\frac{1}{2}\right) = \frac{1}{4},$$

and

$$P(A_2) = P\left(\frac{1}{2} < X < 1\right) = 1 - \frac{1}{4} = \frac{3}{4}.$$

Hence

$$\boxed{P(A_1) = \frac{1}{4}}, \quad \boxed{P(A_2) = \frac{3}{4}}.$$

- (b) Since A_1 and A_2 form a partition of the sample space on the support of X , the total probability theorem gives

$$P(B) = P(B | A_1)P(A_1) + P(B | A_2)P(A_2).$$

Substituting the values,

$$P(B) = \frac{1}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{3}{4} = \frac{1}{16} + \frac{9}{16} = \frac{10}{16} = \frac{5}{8}.$$

Therefore,

$$\boxed{P(B) = \frac{5}{8}}.$$

- (c) By Bayes' theorem,

$$P(A_2 | B) = \frac{P(B | A_2)P(A_2)}{P(B)}.$$

Thus

$$P(A_2 | B) = \frac{\frac{3}{4} \cdot \frac{3}{4}}{\frac{5}{8}} = \frac{9}{16} \cdot \frac{8}{5} = \frac{9}{10}.$$

Hence

$$\boxed{P(A_2 | B) = \frac{9}{10}}.$$

- (d) We have

$$P(\bar{B}) = 1 - P(B) = 1 - \frac{5}{8} = \frac{3}{8}.$$

Also,

$$P(\bar{B} | A_2) = 1 - P(B | A_2) = 1 - \frac{3}{4} = \frac{1}{4}.$$

Therefore, by Bayes' theorem,

$$P(A_2 | \bar{B}) = \frac{P(\bar{B} | A_2)P(A_2)}{P(\bar{B})} = \frac{\frac{1}{4} \cdot \frac{3}{4}}{\frac{3}{8}} = \frac{3}{16} \cdot \frac{8}{3} = \frac{1}{2}.$$

Hence

$$\boxed{P(A_2 | \bar{B}) = \frac{1}{2}}.$$

Question 3 (15 points)

Let (X, Y) be a pair of continuous random variables with joint cumulative distribution function

$$F_{X,Y}(x, y) = \begin{cases} 0, & x \leq 0 \text{ or } y \leq 0, \\ x^2 y, & 0 < x < 1, 0 < y < 1, \\ x^2, & 0 < x < 1, y \geq 1, \\ y, & x \geq 1, 0 < y < 1, \\ 1, & x \geq 1, y \geq 1. \end{cases}$$

- (a) (4 points) Compute the joint pdf $f_{X,Y}(x, y)$ for $0 < x < 1$ and $0 < y < 1$.
- (b) (3 points) Compute the marginal cumulative distribution functions $F_X(x)$ and $F_Y(y)$ for all $x, y \in \mathbb{R}$.
- (c) (3 points) Compute $P\left(X \leq \frac{1}{2}, Y > \frac{1}{4}\right)$.
- (d) (5 points) Are X and Y independent? Justify using the CDFs.

Solution.

- (a) For $0 < x < 1$ and $0 < y < 1$, the joint pdf is obtained by differentiating the joint CDF:

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y).$$

Since $F_{X,Y}(x, y) = x^2 y$ in this region,

$$\frac{\partial}{\partial y} F_{X,Y}(x, y) = x^2, \quad \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y) = 2x.$$

Therefore,

$$\boxed{f_{X,Y}(x, y) = 2x, \quad 0 < x < 1, 0 < y < 1.}$$

- (b) By definition,

$$F_X(x) = F_{X,Y}(x, \infty), \quad F_Y(y) = F_{X,Y}(\infty, y).$$

Thus,

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ x^2, & 0 < x < 1, \\ 1, & x \geq 1, \end{cases}$$

because for $0 < x < 1$ and $y \geq 1$, the joint CDF equals x^2 .

Similarly,

$$F_Y(y) = \begin{cases} 0, & y \leq 0, \\ y, & 0 < y < 1, \\ 1, & y \geq 1, \end{cases}$$

because for $x \geq 1$ and $0 < y < 1$, the joint CDF equals y .

Hence

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ x^2, & 0 < x < 1, \\ 1, & x \geq 1, \end{cases} \quad F_Y(y) = \begin{cases} 0, & y \leq 0, \\ y, & 0 < y < 1, \\ 1, & y \geq 1. \end{cases}$$

(c) We use

$$P\left(X \leq \frac{1}{2}, Y > \frac{1}{4}\right) = P\left(X \leq \frac{1}{2}\right) - P\left(X \leq \frac{1}{2}, Y \leq \frac{1}{4}\right).$$

From part (b),

$$P\left(X \leq \frac{1}{2}\right) = F_X\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}.$$

Also,

$$P\left(X \leq \frac{1}{2}, Y \leq \frac{1}{4}\right) = F_{X,Y}\left(\frac{1}{2}, \frac{1}{4}\right) = \left(\frac{1}{2}\right)^2 \left(\frac{1}{4}\right) = \frac{1}{16}.$$

Therefore,

$$P\left(X \leq \frac{1}{2}, Y > \frac{1}{4}\right) = \frac{1}{4} - \frac{1}{16} = \frac{3}{16}.$$

Hence

$$P\left(X \leq \frac{1}{2}, Y > \frac{1}{4}\right) = \frac{3}{16}.$$

(d) From part (b),

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ x^2, & 0 < x < 1, \\ 1, & x \geq 1, \end{cases} \quad F_Y(y) = \begin{cases} 0, & y \leq 0, \\ y, & 0 < y < 1, \\ 1, & y \geq 1. \end{cases}$$

We compare $F_{X,Y}(x, y)$ with $F_X(x)F_Y(y)$.

If $x \leq 0$ or $y \leq 0$, then both are 0.

If $0 < x < 1$ and $0 < y < 1$, then

$$F_X(x)F_Y(y) = x^2y = F_{X,Y}(x, y).$$

If $0 < x < 1$ and $y \geq 1$, then

$$F_X(x)F_Y(y) = x^2 \cdot 1 = x^2 = F_{X,Y}(x, y).$$

If $x \geq 1$ and $0 < y < 1$, then

$$F_X(x)F_Y(y) = 1 \cdot y = y = F_{X,Y}(x, y).$$

If $x \geq 1$ and $y \geq 1$, then

$$F_X(x)F_Y(y) = 1 \cdot 1 = 1 = F_{X,Y}(x, y).$$

Therefore,

$$F_{X,Y}(x, y) = F_X(x)F_Y(y)$$

for all $x, y \in \mathbb{R}$, so X and Y are independent.

Question 4 (20 points)

Let X and Y be continuous random variables with joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} kxy, & 0 < x < 1, 0 < y < 2x, \\ 0, & \text{otherwise,} \end{cases}$$

where $k > 0$ is a normalizing constant.

- (a) (5 points) Determine the value of k so that $f_{X,Y}$ is a valid joint pdf.
- (b) (5 points) Compute the marginal pdfs $f_X(x)$ and $f_Y(y)$.
- (c) (5 points) Compute the conditional pdf $f_{Y|X}(y|x)$.
- (d) (5 points) Compute $P\left(Y > 1 \mid X = \frac{3}{4}\right)$.

Solution.

- (a) For $f_{X,Y}$ to be a valid joint pdf, we need:

$$\int_0^1 \int_0^{2x} kxy \, dy \, dx = 1.$$

Compute the inner integral:

$$\int_0^{2x} kxy \, dy = kx \cdot \frac{y^2}{2} \Big|_0^{2x} = kx \cdot \frac{(2x)^2}{2} = kx \cdot 2x^2 = 2kx^3.$$

Now the outer integral:

$$\int_0^1 2kx^3 \, dx = 2k \cdot \frac{x^4}{4} \Big|_0^1 = 2k \cdot \frac{1}{4} = \frac{k}{2}.$$

Setting $\frac{k}{2} = 1$ gives $\boxed{k = 2}$.

- (b) **Marginal of X :** For $0 < x < 1$,

$$f_X(x) = \int_0^{2x} 2xy \, dy = 2x \cdot \frac{y^2}{2} \Big|_0^{2x} = 2x \cdot 2x^2 = 4x^3.$$

Thus $f_X(x) = 4x^3$ for $0 < x < 1$, and 0 otherwise.

Marginal of Y : The support region is $0 < x < 1$, $0 < y < 2x$. For a given y , the constraint $y < 2x$ gives $x > y/2$. Also $x < 1$, so x ranges from $y/2$ to 1. Since $y < 2x \leq 2$, the marginal is defined for $0 < y < 2$:

$$f_Y(y) = \int_{y/2}^1 2xy \, dx = 2y \cdot \frac{x^2}{2} \Big|_{y/2}^1 = y \left(1 - \frac{y^2}{4}\right) = y - \frac{y^3}{4}.$$

Thus $f_Y(y) = y - \frac{y^3}{4}$ for $0 < y < 2$, and 0 otherwise.

- (c) By definition:

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{2xy}{4x^3} = \frac{y}{2x^2}, \quad 0 < y < 2x.$$

We can verify: $\int_0^{2x} \frac{y}{2x^2} \, dy = \frac{1}{2x^2} \cdot \frac{(2x)^2}{2} = \frac{4x^2}{4x^2} = 1. \checkmark$

(d) With $x = 3/4$, the conditional pdf is:

$$f_{Y|X}\left(y \mid \frac{3}{4}\right) = \frac{y}{2(3/4)^2} = \frac{y}{2 \cdot 9/16} = \frac{8y}{9}, \quad 0 < y < \frac{3}{2}.$$

Therefore:

$$P\left(Y > 1 \mid X = \frac{3}{4}\right) = \int_1^{3/2} \frac{8y}{9} dy = \frac{8}{9} \cdot \frac{y^2}{2} \Big|_1^{3/2} = \frac{8}{9} \cdot \frac{1}{2} \left(\frac{9}{4} - 1\right) = \frac{8}{9} \cdot \frac{1}{2} \cdot \frac{5}{4} = \frac{8 \cdot 5}{9 \cdot 8} = \frac{5}{9}.$$

Question 5 (15 points)

Let X be a continuous random variable with pdf

$$f_X(x) = \begin{cases} 3x^2, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$A = \left\{ \frac{1}{2} < X < 1 \right\}.$$

(a) (4 points) Compute $P(A)$.

(b) (6 points) Compute the conditional cumulative distribution function

$$F_{X|A}(x|A) = P(X \leq x \mid A)$$

for all $x \in \mathbb{R}$.

(c) (5 points) Use your result from part (b) to compute $P\left(X > \frac{3}{4} \mid A\right)$.

Solution.

(a) Since

$$F_X(x) = \int_0^x 3t^2 dt = x^3, \quad 0 < x < 1,$$

we have

$$P(A) = P\left(\frac{1}{2} < X < 1\right) = F_X(1) - F_X\left(\frac{1}{2}\right) = 1 - \frac{1}{8} = \frac{7}{8}.$$

Therefore,

$$\boxed{P(A) = \frac{7}{8}}.$$

(b) By definition,

$$F_{X|A}(x|A) = P(X \leq x \mid A) = \frac{P(X \leq x, A)}{P(A)}.$$

We consider cases:

If $x \leq \frac{1}{2}$, then $P(X \leq x, A) = 0$, so

$$F_{X|A}(x|A) = 0.$$

If $\frac{1}{2} < x < 1$, then

$$\{X \leq x, A\} = \left\{ \frac{1}{2} < X \leq x \right\},$$

and hence

$$P(X \leq x, A) = F_X(x) - F_X\left(\frac{1}{2}\right) = x^3 - \frac{1}{8}.$$

Therefore,

$$F_{X|A}(x|A) = \frac{x^3 - \frac{1}{8}}{\frac{7}{8}} = \frac{8x^3 - 1}{7}.$$

If $x \geq 1$, then $F_{X|A}(x|A) = 1$.

Thus

$$F_{X|A}(x|A) = \begin{cases} 0, & x \leq \frac{1}{2}, \\ \frac{8x^3 - 1}{7}, & \frac{1}{2} < x < 1, \\ 1, & x \geq 1. \end{cases}$$

(c) From part (b),

$$P\left(X > \frac{3}{4} \mid A\right) = 1 - F_{X|A}\left(\frac{3}{4} \mid A\right).$$

Since $\frac{1}{2} < \frac{3}{4} < 1$,

$$F_{X|A}\left(\frac{3}{4} \mid A\right) = \frac{8\left(\frac{3}{4}\right)^3 - 1}{7} = \frac{8 \cdot \frac{27}{64} - 1}{7} = \frac{\frac{27}{8} - 1}{7} = \frac{19}{56}.$$

Therefore,

$$P\left(X > \frac{3}{4} \mid A\right) = 1 - \frac{19}{56} = \frac{37}{56}.$$

Question 6 (15 points)

Let Θ be a continuous random variable with prior density

$$f_{\Theta}(\theta) = \begin{cases} 2\theta, & 0 < \theta < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Given $\Theta = \theta$, the observation X has conditional density

$$f_{X|\Theta}(x|\theta) = \begin{cases} \frac{1}{\theta}, & 0 < x < \theta, \\ 0, & \text{otherwise.} \end{cases}$$

(a) (6 points) Compute the evidence (marginal density of X):

$$f_X(x) = \int_0^1 f_{X|\Theta}(x|\theta) f_{\Theta}(\theta) d\theta, \quad 0 < x < 1.$$

(b) (9 points) Using Bayes' theorem, derive the posterior density $f_{\Theta|X}(\theta|x)$ for $x < \theta < 1$ and $0 < x < 1$.

Solution.

(a) For $0 < x < 1$, we start from

$$f_X(x) = \int_0^1 f_{X|\Theta}(x|\theta) f_{\Theta}(\theta) d\theta.$$

Since

$$f_{X|\Theta}(x|\theta) = 0 \quad \text{when } \theta \leq x,$$

only values $x < \theta < 1$ contribute to the integral. Therefore,

$$\begin{aligned} f_X(x) &= \int_x^1 \frac{1}{\theta} \cdot 2\theta \, d\theta \\ &= \int_x^1 2 \, d\theta \\ &= 2(1-x). \end{aligned}$$

Thus

$$\boxed{f_X(x) = 2(1-x), \quad 0 < x < 1.}$$

(b) By Bayes' theorem for continuous random variables:

$$f_{\Theta|X}(\theta|x) = \frac{f_{X|\Theta}(x|\theta) f_{\Theta}(\theta)}{f_X(x)}, \quad x < \theta < 1.$$

Using part (a),

$$f_{\Theta|X}(\theta|x) = \frac{\frac{1}{\theta} \cdot 2\theta}{2(1-x)} = \frac{1}{1-x}, \quad x < \theta < 1.$$

Therefore,

$$\boxed{f_{\Theta|X}(\theta|x) = \begin{cases} \frac{1}{1-x}, & x < \theta < 1, \\ 0, & \text{otherwise,} \end{cases} \quad 0 < x < 1.}$$

This is the density of a uniform distribution on the interval $(x, 1)$.

Bonus (10 points)

Let X and Y be continuous random variables with joint pdf $f_{X,Y}(x,y)$. Assume that the marginal pdf $f_Y(y)$ is positive for the value of y under consideration, and that the conditional pdf $f_{Y|X}(y|x)$ is well-defined.

(a) (5 points) Show that

$$f_Y(y) = \int_{-\infty}^{\infty} f_{Y|X}(y|x) f_X(x) \, dx.$$

(b) (5 points) Use part (a) and the product rule to derive the continuous Bayes formula

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x) f_X(x)}{\int_{-\infty}^{\infty} f_{Y|X}(y|u) f_X(u) \, du}.$$

Solution.

(a) By the product rule for continuous random variables,

$$f_{X,Y}(x,y) = f_{Y|X}(y|x) f_X(x).$$

Now integrate both sides with respect to x over $\mathbb{R}$:

$$\int_{-\infty}^{\infty} f_{X,Y}(x,y) \, dx = \int_{-\infty}^{\infty} f_{Y|X}(y|x) f_X(x) \, dx.$$

By definition, the left-hand side is the marginal pdf $f_Y(y)$. Therefore,

$$\boxed{f_Y(y) = \int_{-\infty}^{\infty} f_{Y|X}(y|x) f_X(x) \, dx.}$$

(b) Using the product rule in the other direction,

$$f_{X,Y}(x,y) = f_{X|Y}(x|y) f_Y(y).$$

Combining this with

$$f_{X,Y}(x,y) = f_{Y|X}(y|x) f_X(x),$$

we obtain

$$f_{X|Y}(x|y) f_Y(y) = f_{Y|X}(y|x) f_X(x).$$

Since $f_Y(y) > 0$, divide by $f_Y(y)$:

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x) f_X(x)}{f_Y(y)}.$$

Now substitute the expression for $f_Y(y)$ from part (a):

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x) f_X(x)}{\int_{-\infty}^{\infty} f_{Y|X}(y|u) f_X(u) du}.$$

This is the continuous Bayes formula.